Modeling and Validation of a Soccer Ball's Three-Dimensional Flight to the Goal

Problem B

Abstract

A penalty kick reaches the goal quickly and can follow a curved path; accurate position prediction can help a goalkeeper judge its direction in time. Using the ball's initial position and early position observations, we establish and compare four motion models that respectively account for gravity, drag, a constant resultant force, and Magnus force. The comprehensive model incorporates gravity, drag, and Magnus force, with a constant drag coefficient and a linear lift relation. Its initial motion state and aerodynamic parameters are inferred from early positions, and the subsequent trajectory is obtained by numerical integration.

For the positions given in the problem, the comprehensive model predicts that the ball reaches the goal plane at approximately 0.399 s, with its center at $(11,\ 2.616,\ 1.953)$ m. To evaluate prediction performance, we select trajectories from a real penalty-kick dataset that match the problem setting and divide their positions into conditioning and validation points. The former are used for parameter estimation and the latter for evaluating predictions. We consider two types of shot, accurate placement shots and powerful driven shots, and use the measured trajectories to examine their motion characteristics and prediction performance, assessing the physical accuracy of the comprehensive model.

Sensitivity analysis shows that the initial direction has a substantial effect on the goal-plane crossing: a change of 1° shifts the crossing by approximately 0.20 m. We also analyze the range of crossing positions arising from uncertainty in the initial motion state and aerodynamic parameters. These results can support trajectory prediction and a goalkeeper's judgment of the incoming direction.

Keywords: penalty trajectory; motion model; Magnus force; parameter estimation; uncertainty analysis

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1 Introduction

1.1 Problem Background

A penalty kick is a direct contest between a designated kicker and the opposing goalkeeper. The kicker strikes the ball from the penalty mark towards the goal. In a typical professional penalty, the ball then flies freely through the air towards the goal, while the goalkeeper must choose where and when to move. An effective shot depends on speed and placement, as well as the kicker's ability to conceal its intended direction.

The importance of penalties to match outcomes has motivated research into shooting and saving, particularly the ball's motion after it leaves the foot. Topics include curved trajectories and aerodynamic forces[3], surface structure[5, 4], and trajectory measurements and aerodynamic-parameter estimation[9]. These studies connect kicking technique with the physics of ball flight and help explain the motion of a penalty shot.

1.2 Problem Restatement

Problem B concerns the prediction of a penalty trajectory from its early flight positions. The goal is 11 m from the penalty mark, 7.32 m wide and 2.44 m high. The task is to develop and test a general method that uses the ball-centre coordinates at 0.05 s and 0.10 s after the kick to predict its most likely entry point and estimate the uncertainty of that prediction.

The preceding discussion suggests three broad stages of the ball's motion: **contact with the kicker**, **free airborne flight** and **motion after the shot**. We examine the free flight after ball–foot contact has ended and before the ball enters the goal; the effect of contact is represented by the initial velocity and spin. The forces and governing equations are discussed further in Section 3. Initial-state inference from early positions, subsequent trajectory prediction and assessment of those predictions are discussed further in Section 4. Section 5 uses the specified example to examine parameter sensitivity and the distribution of goal-plane crossings under stated perturbations, connecting the predicted location with its uncertainty. The technical workflow is summarized in Figure 1.

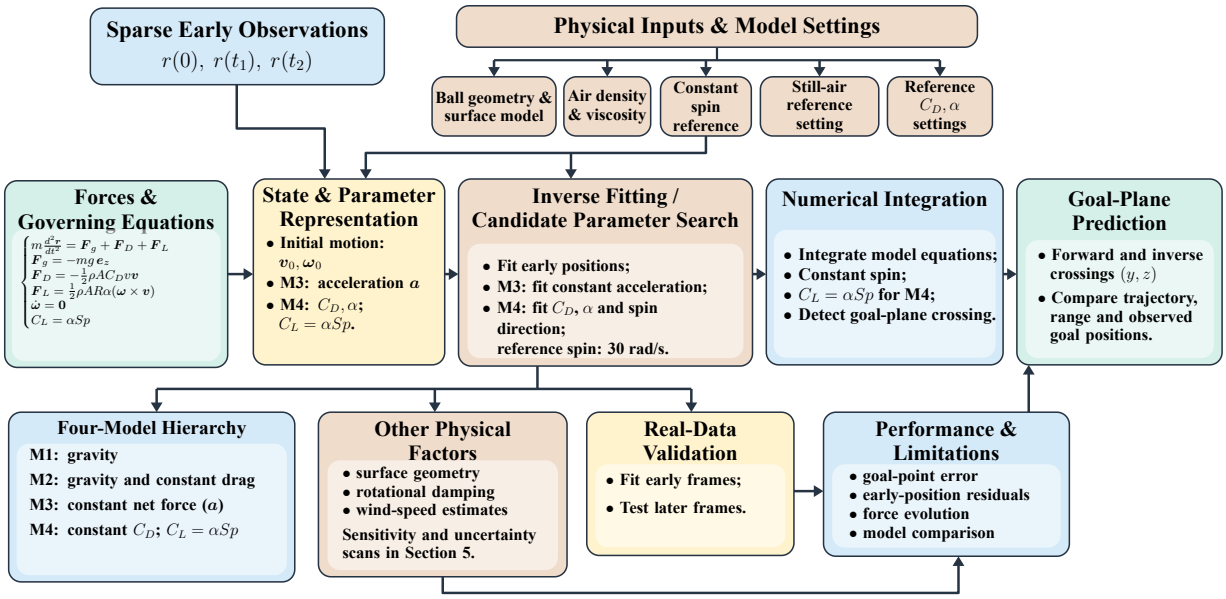


Figure 1: Three-force modeling, inverse prediction and real-penalty validation.

2 Preparation of the Models

2.1 Assumptions

1. **Free flight.** At the start of the motion considered, the ball has left the kicker's foot; contact is represented by the initial velocity and spin. We assume a proficient kick, with the ball remaining airborne without contacting the ground or another object before reaching the goal plane.

2. **Rigid spherical ball.** The ball remains spherical and its mass is uniformly distributed. Seams and asymmetric surface geometry are discussed further in Section 6.1.

3. **Uniform gravity and air properties.** The magnitude and direction of gravitational acceleration remain constant over the flight, and air density and viscosity remain uniform.

4. **Still air.** The surrounding air is treated as stationary. The effects of wind on relative airflow and aerodynamic force are discussed further in Section 6.3.

5. **Constant spin.** Aerodynamic torque is neglected, so the spin vector remains constant during flight. Rotational damping is discussed further in Section 6.2.

2.2 Notations

The symbols used below and their physical meanings are summarized in Table 1.

Table 1: Principal quantities and competition reference values.

Symbol	Meaning	Value or unit
$\boldsymbol{r}, \boldsymbol{v}$	Ball-centre position and velocity	m; m/s
$\boldsymbol{\omega}, \omega$	Spin vector and its magnitude	rad/s
m, R, A	Mass, radius and frontal area	0.43 kg; 0.11 m; πR^2
g	Gravitational acceleration	9.81 m/s^2
ρ	Air density	1.2 kg/m^3
$\nu, \mu = \rho\nu$	Kinematic and dynamic viscosity	1.50×10^{-5} m^2/s; Pa s
Re, Sp	Reynolds number and spin parameter	dimensionless
C_D, C_L, α	Drag, lift and linear-spin parameters	dimensionless
$\lambda_D, \lambda_\alpha$	Coefficient ratios to reference values	dimensionless
$\theta, \boldsymbol{e}_L$	Spin–airflow angle and Magnus direction	rad; unit vector
q, v	Dynamic pressure and ball speed	Pa; m/s
$\boldsymbol{Q}, I_b$	Damping torque and moment of inertia	N m; kg m^2

Ball and air parameters are taken from prior studies [3, 5], with dynamic viscosity $\mu = \rho\nu$.

3 Model Establishment

3.1 Geometric Relations

As shown in Figure 2, the local origin is the penalty mark, with x towards the goal, y horizontal and lateral, and z vertically upward. The goal plane is $x = 11$ m and the opening is $-3.66 \le y \le 3.66$ m, $0 \le z \le 2.44$ m.

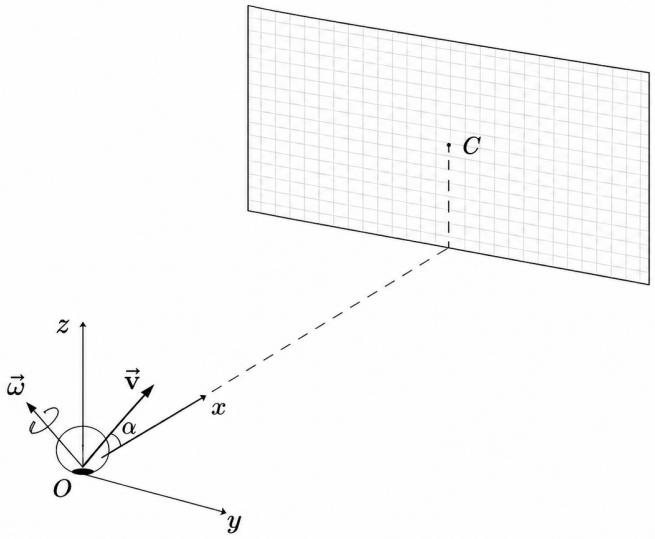


Figure 2: Penalty-kick geometry in local coordinates; O is the nominal penalty mark and C an illustrative goal-plane point.

The entire ball clears the opening when its centre satisfies $|y| \leq 3.66 - R$ and $R \leq z \leq 2.44 - R$.

3.2 Force Analysis

As shown in Figure 3, we consider gravity, buoyancy, drag and Magnus force in turn.

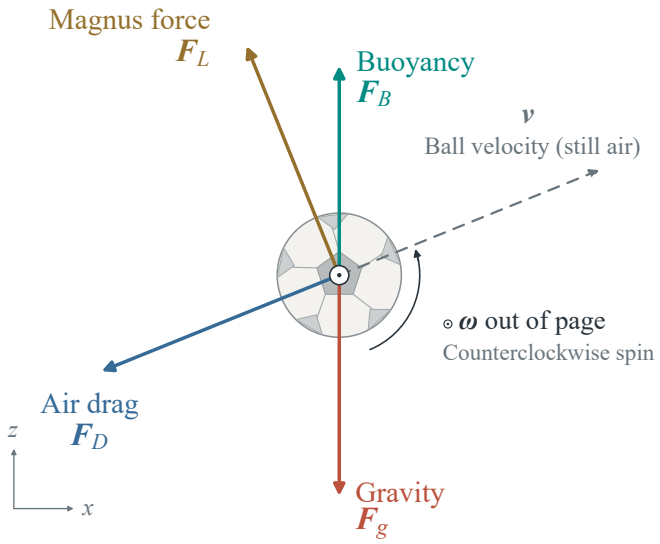


Figure 3: Schematic of gravity, buoyancy, drag and Magnus force on the football.

3.2.1 Gravity

Let $\boldsymbol{e}_z$ be the upward vertical unit vector. For mass m in the uniform field of gravitational acceleration magnitude g,

$$\boldsymbol{F}_g = -mg\boldsymbol{e}_z, \tag{1}$$

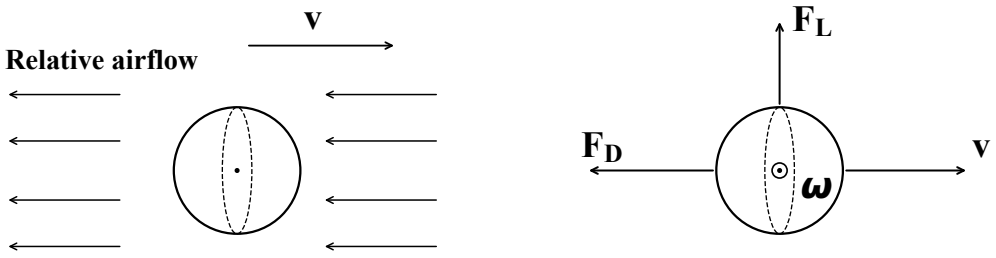
By Archimedes' principle, the displaced air also exerts an upward buoyancy force $\boldsymbol{F}_B$. For ball volume $V_b = \frac{4}{3}\pi R^3$ and air density ρ,

$$\boldsymbol{F}_B = \rho V_b g\, \boldsymbol{e}_z = \frac{4}{3}\pi R^3 \rho g\, \boldsymbol{e}_z. \tag{2}$$

From these expressions, the combined vertical force is $\boldsymbol{F}_g + \boldsymbol{F}_B = -(m - \rho V_b)g\boldsymbol{e}_z$. Buoyancy is 1.56% of weight for the reference parameters and is therefore omitted as a small correction in the subsequent calculations.

3.2.2 Aerodynamic Drag

Let $\boldsymbol{v}$ be the ball-centre velocity. Since the air is assumed stationary, the incident airflow in the ball frame has velocity $-\boldsymbol{v}$. As shown in Figure 4, the airflow slows near the forward face of the ball.



(a) Airflow in the ball frame **(b) Example: spin out of the page**

Figure 4: Schematic of relative airflow and force directions.

Let p_∞ be the incident static pressure and p_s the pressure at the forward stagnation point. Bernoulli's relation along an idealized incident streamline is

$$p_\infty + \frac{1}{2}\rho v^2 = p_s.$$

The pressure rise defines the dynamic pressure q:

$$q = p_s - p_\infty = \frac{1}{2}\rho v^2.$$

Following an established aerodynamic model [9], aerodynamic drag depends on air density, relative speed, frontal area and the ball's flow and surface characteristics. The latter enter through the drag coefficient C_D, so the drag magnitude is

$$F_D = C_D q A.$$

Since drag opposes the relative motion, substitution of the dynamic pressure leads to

$$\boldsymbol{F}_D = -\frac{1}{2}\rho A C_D v \boldsymbol{v}. \tag{3}$$

Here $A = \pi R^2$ is the frontal area and ρ is air density. We approximate C_D as constant during the ball's flight.

3.2.3 Magnus Lift

A spinning football changes the surrounding airflow and experiences a transverse Magnus force. As shown in Figure 5, rotation makes the relative flow asymmetric. Let $\boldsymbol{\omega}$ be the angular-velocity vector.

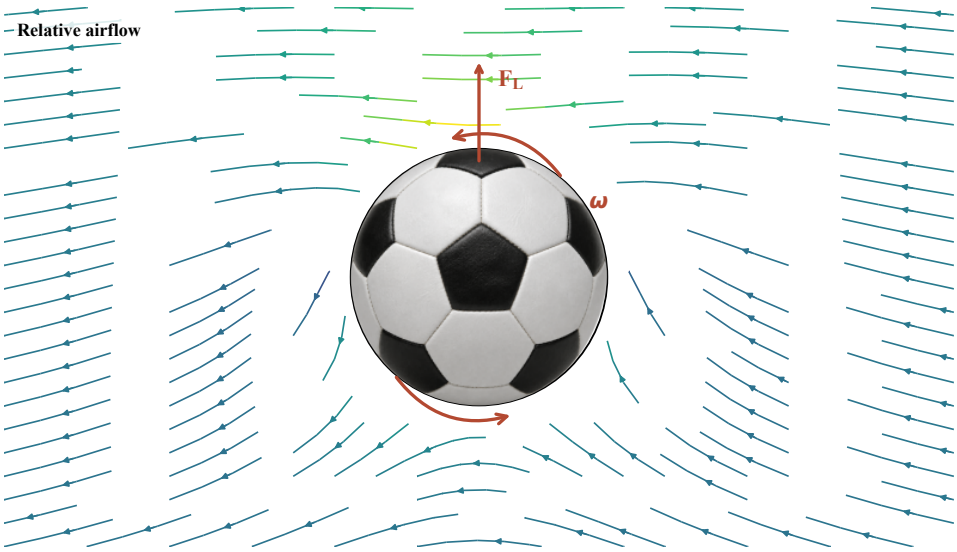


Figure 5: Idealized flow around a spinning football.

Following an established aerodynamic model [9], transverse lift is expressed in terms of dynamic pressure and frontal area. When the spin axis makes angle θ with the relative velocity, the lift magnitude is

$$F_L = qAC_L \sin\theta.$$

Here C_L is the transverse lift coefficient and θ is the spin–airflow angle. We approximate the relation between lift coefficient and spin parameter as linear:

$$C_L = \alpha \frac{R\omega}{v}.$$

Here α is dimensionless. Following prior work [6], we approximate this coefficient as constant during the flight. Substituting the dynamic-pressure and linear-spin relations,

$$\boldsymbol{F}_L = \frac{1}{2}\rho A R \alpha (\boldsymbol{\omega} \times \boldsymbol{v}). \tag{4}$$

3.3 Dynamical Equations

Since the football is treated as a rigid body, its motion can be separated into translation of the centre of mass and rotation about that centre. Let $\boldsymbol{L}$ be angular momentum about the centre, I_b the moment of inertia, and $\boldsymbol{Q}$ the resultant external torque. Centre-of-mass translation follows Newton's second law,

$$m\dot{\boldsymbol{v}} = \sum \boldsymbol{F} \tag{5}$$

while rotation about the centre follows the angular-momentum theorem,

$$\frac{d\boldsymbol{L}}{dt} = \boldsymbol{Q}, \qquad \boldsymbol{L} = I_b\boldsymbol{\omega}. \tag{6}$$

From zero applied torque and constant spherical inertia, $I_b\dot{\boldsymbol{\omega}} = \boldsymbol{0}$, so the spin remains constant. Combining this result with the three retained forces,

$$\begin{cases} \dot{\boldsymbol{r}} = \boldsymbol{v}, \\ m\dot{\boldsymbol{v}} = \boldsymbol{F}_g + \boldsymbol{F}_D + \boldsymbol{F}_L, \\ \dot{\boldsymbol{\omega}} = \boldsymbol{0}. \end{cases} \tag{7}$$

Let $\boldsymbol{r}_0$, $\boldsymbol{v}_0$ and $\boldsymbol{\omega}_0$ be the initial centre position, velocity and angular velocity, respectively; their subscripts x, y, z denote components in the chosen axes. The initial conditions are

$$\begin{cases} \boldsymbol{r}(0) = \boldsymbol{r}_0, \\ \boldsymbol{v}(0) = (v_{x0}, v_{y0}, v_{z0}), \\ \boldsymbol{\omega}(0) = (\omega_{x0}, \omega_{y0}, \omega_{z0}). \end{cases} \tag{8}$$

4 Model Solving

This chapter develops models and an algorithm for inferring the initial motion from early ball positions and predicting the goal-plane crossing.

4.1 Solution Method

4.1.1 Model selection

Based on the force analysis in Chapter 3, we compare four models with different force assumptions:

1. **M1: gravity.** With $\boldsymbol{a} = (0, 0, -g)$, only the three components of $\boldsymbol{v}_0$ are estimated. Spin has no effect on the trajectory.

2. **M2: gravity and constant-coefficient drag.** Only weight and drag opposite to velocity are considered. The estimated parameters are the initial velocity and constant C_D, for four unknowns in total.

3. **M3: constant resultant force.** All forces are represented by a constant resultant acceleration $\boldsymbol{a}$. The three components of $\boldsymbol{v}_0$ and three of $\boldsymbol{a}$ form six unknowns. Gravity is included in this resultant.

4. **M4: gravity, drag and Magnus force.** The force relations of Section 3 determine the motion. We estimate three initial velocity components, two spin-direction angles, C_D and the linear lift parameter α: seven unknowns.

4.1.2 Inverse method

For measured positions $\boldsymbol{r}_i$ at times t_i ($i = 0, 1, 2$), let $\widehat{\boldsymbol{r}}(t_i; \boldsymbol{p})$ denote the prediction associated with the unknown parameter vector $\boldsymbol{p}$. We minimize the six coordinate residuals through

$$J(\boldsymbol{p}) = \sum_{i=1}^{2} \left\| \widehat{\boldsymbol{r}}(t_i; \boldsymbol{p}) - \boldsymbol{r}_i \right\|^2 .$$

The parameter bounds for the four models appear in Appendix A.

After fitting, the inferred state is integrated to the goal plane $x = 11$ m. The goal-crossing distance error is

$$E_g = \sqrt{(\widehat{y}_g - y_g)^2 + (\widehat{z}_g - z_g)^2}, \tag{9}$$

where (y_g, z_g) is the observed crossing and hats denote the prediction. Further parameter and source records appear in Appendix A.

4.1.3 Dataset

To evaluate prediction performance, we select three trajectories from a real penalty-kick dataset[1, 2], referred to as Cases 1, 2 and 3. They include two placement shots and one power shot. The data sources are described in Appendix A.

The selected starting positions are $\boldsymbol{r}_0 = (3.04, 0.55, 0.13)$ m, $(2.57, 0.85, 0.34)$ m and $(0.77, 0.08, 0.24)$ m for Case 1, Case 2 and Case 3, respectively; each defines $t = 0$. As shown in Table 2, the next two positions occur at 0.04 and 0.08 s.

Table 2: Two later input positions and observed goal crossings; coordinates in m.

Case	t (s)	(x, y, z)	Observed (y_g, z_g)
Case 1	0.04	$(4.19, 0.82, 0.15)$	$(2.7500, 0.1600)$
Case 1	0.08	$(5.25, 1.07, 0.18)$	
Case 2	0.04	$(3.64, 1.20, 0.41)$	$(3.0697, 0.4657)$
Case 2	0.08	$(4.71, 1.53, 0.47)$	
Case 3	0.04	$(1.99, 0.34, 0.46)$	$(2.9188, 1.1789)$
Case 3	0.08	$(3.18, 0.62, 0.65)$	

4.2 Case Analysis

Prior research shows [11] that penalty technique and ball speed affect accuracy and the goalkeeper's anticipation. Studying the kicker's technique helps interpret and predict penalty trajectories. We distinguish two common approaches:

1. **Precise placement shot.** A side-foot contact favours control of direction and accurate placement. A low trajectory can carry the ball towards a lateral target while keeping it close to the turf.

2. **Power instep shot.** An instep contact emphasizes a fast strike, shortening the goalkeeper's available response time. The launch direction can produce a higher flight towards the goal.

Reviewing the match footage, we classify Case 1 and Case 2 as placement shots and Case 3 as a power shot. As shown in Figure 6, Figures 6a and 6b illustrate the low placement and higher power-shot flight patterns, respectively.



(a) Illustrative placement shot.



(b) Illustrative power shot.

Figure 6: Photographic examples of placement and power shots.

4.2.1 Precise placement shots

As shown in Figure 7, both placement shots remain low.

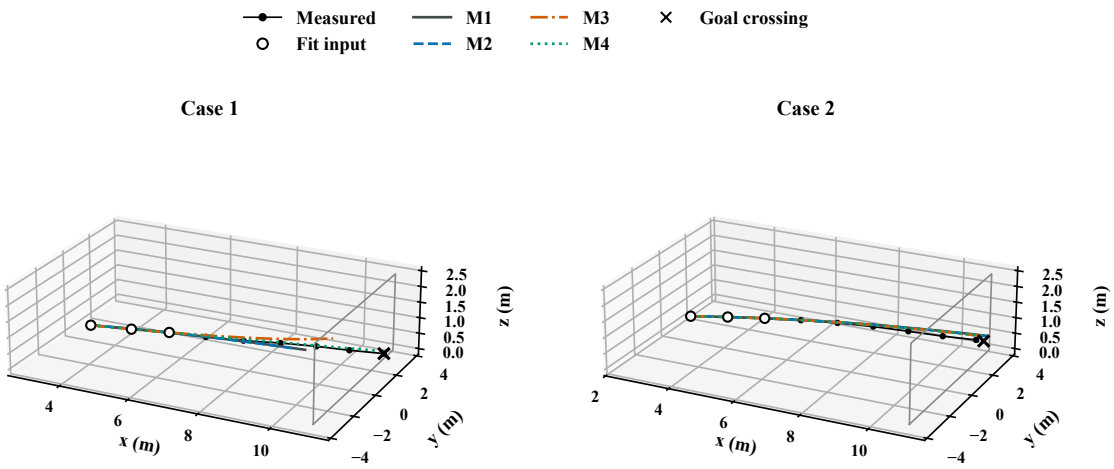


Figure 7: Measured and predicted trajectories for the two placement shots.

In M4, the initial vertical velocities are 0.55 and 1.89 m/s for the two cases. As shown in Figures 7 and 8, M1 and M2 touch the ground early in Case 1, and M3 does not reach the goal plane, while M4 stays closer to the measured flight. Comparing the four models suggests that the upward Magnus force associated with spin partly offsets gravitational fall and allows the ball to continue towards the goal along a low trajectory. In Case 2, both M3 and M4 reproduce bending in the y direction, with M4 providing the most accurate prediction. Thus, the small upward velocity keeps a placement shot low, while Magnus force plays a key role in maintaining its height.

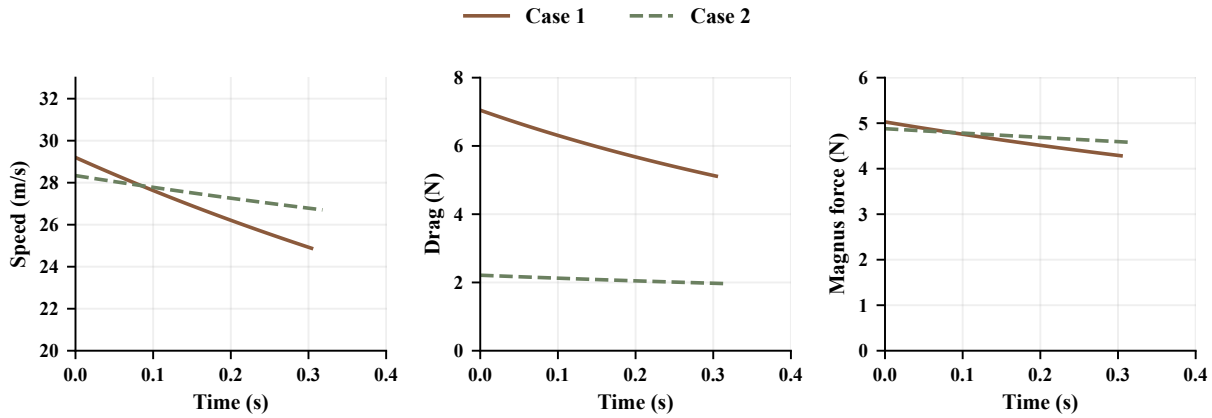


Figure 8: M4 speed, drag and Magnus-force magnitude for the two placement shots. Axis scales match Figure 10.

4.2.2 Power instep shot

As shown in Figure 9, Case 3 rises substantially higher than the two placement shots.

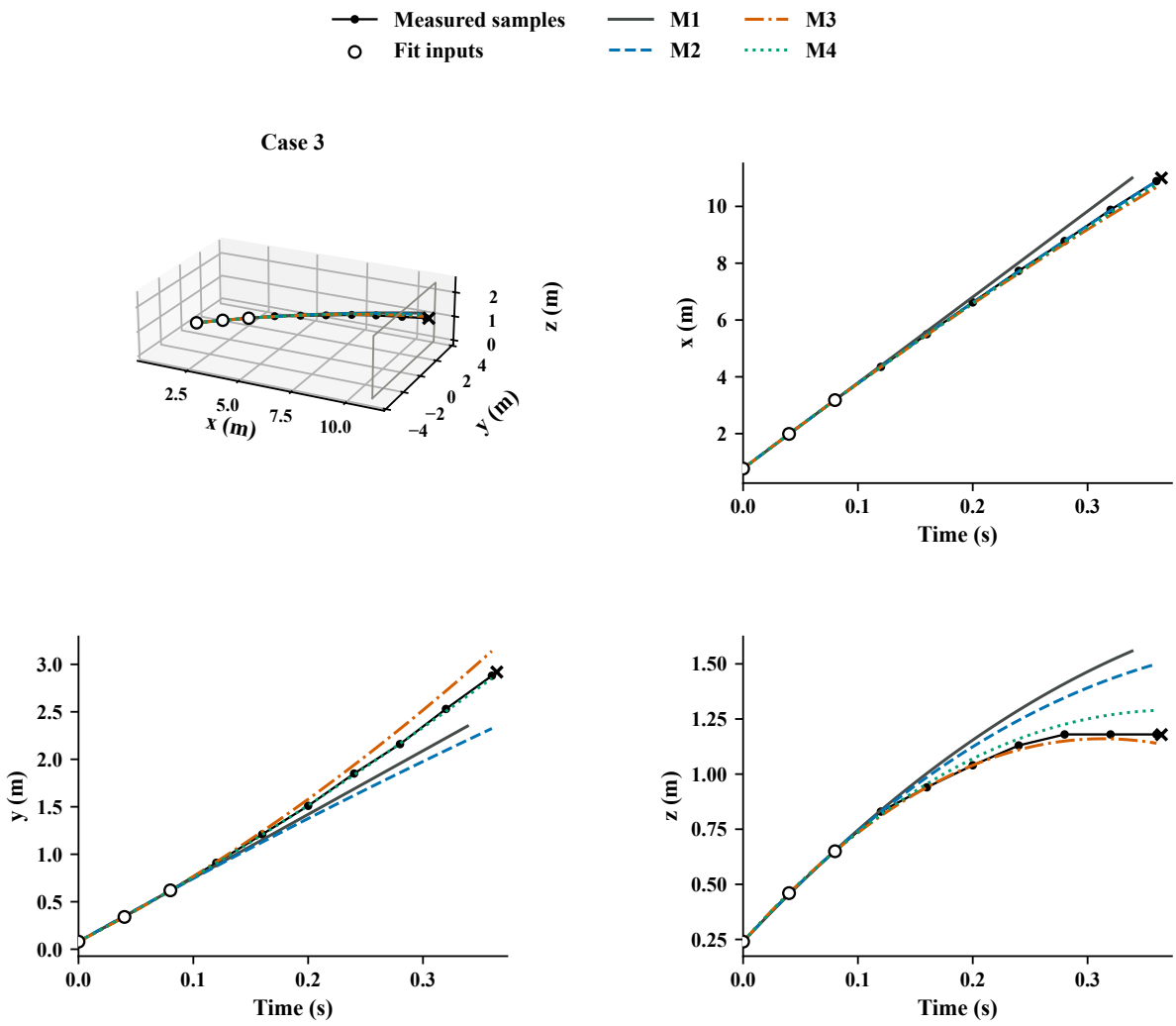


Figure 9: Case 3: spatial trajectory, $x(t)$, $y(t)$ and $z(t)$ in a 2×2 layout. Symbols follow Figure 7.

The M4 initial speed is 32.05 m/s, exceeding the two placement values of 29.20 and 28.34 m/s. Its elevation is 10.46° and initial vertical velocity 5.82 m/s, producing a higher flight. As shown in Figure 10 and Table 3, M1 and M2 do not adequately capture the lateral bending. The M3 and M4 trajectories show how the upward initial velocity raises the ball, while gravity and the downward Magnus component in M4 bend the flight back. Magnus force therefore plays a key role in the lateral curvature of this power shot.

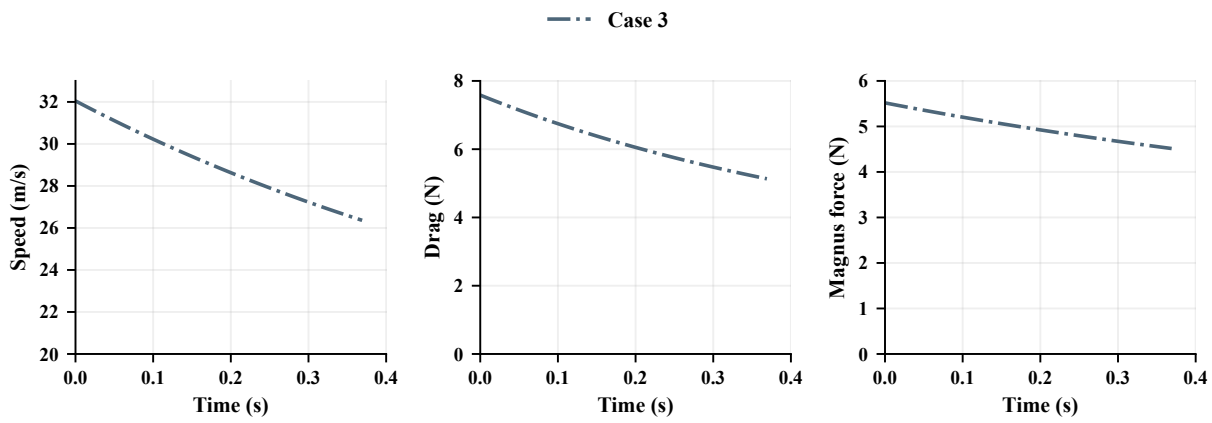


Figure 10: M4 speed, drag and Magnus-force magnitude for the power shot, on the same axes as Figure 8.

Table 3: Power-shot goal crossings and distance errors (m).

Case	Model	Observed (y, z)	Predicted (y, z)	Distance
Case 3	M1	$(2.919, 1.179)$	$(2.350, 1.558)$	0.684
Case 3	M2	$(2.919, 1.179)$	$(2.349, 1.506)$	0.657
Case 3	M3	$(2.919, 1.179)$	$(3.289, 1.126)$	0.374
Case 3	M4	$(2.919, 1.179)$	$(2.934, 1.291)$	0.113

Considering both a valid goal-plane prediction and its distance error, accounting for drag and spin, especially the changing Magnus force during flight, makes **M4 the better model for these three cases.**

4.3 M4 Inversion for the Problem B Observations

The initial centre is $(0, 0, 0.11)$ m. As shown in Table 4, Problem B specifies two later positions at 0.05 and 0.10 s. We apply the same M4 inverse method and parameter settings to these inputs.

Table 4: Initial position and two observations specified in Problem B.

t (s)	x (m)	y (m)	z (m)
0	0	0	0.11
0.05	1.50	0.45	0.44
0.10	2.96	0.86	0.74

The inferred initial motion is

$$\begin{aligned} \boldsymbol{v}_0 &= (30.3784, 9.4074, 6.8974)\ \mathrm{m/s}, \\ \boldsymbol{\omega}_0 &= (19.7434, 1.2060, -22.5553)\ \mathrm{rad/s}. \end{aligned} \tag{10}$$

As shown in Figure 11, **the predicted goal-plane crossing is** $(y_g, z_g) = (2.6159, 1.9526)$ **m**. The inferred speed of 32.54 m/s, elevation of 12.24° and pronounced lateral bending indicate that **this trajectory more closely resembles the power-shot type studied in this chapter.**

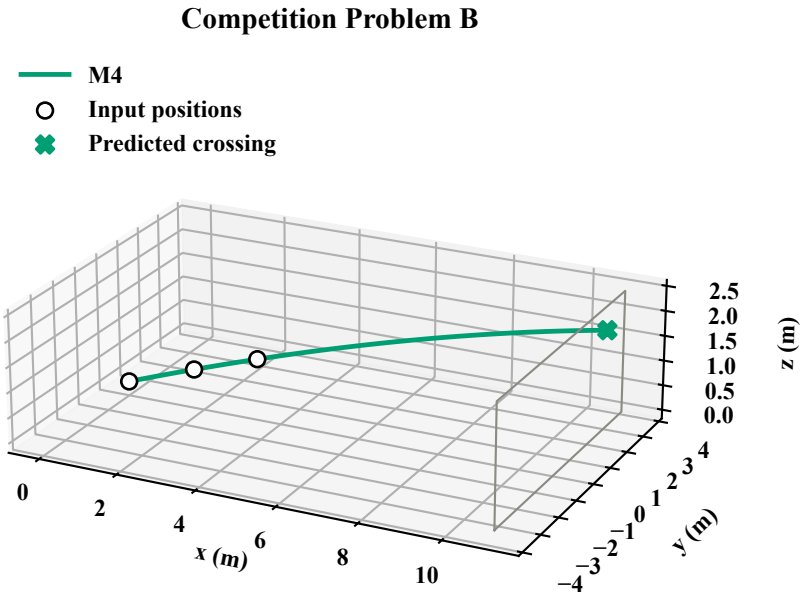


Figure 11: M4 prediction for Problem B, showing the three input positions, goal and predicted crossing.

Section 5 uses this inferred state to study changes in the goal crossing and its distribution.

5 Sensitivity Analysis

Changes in launch speed, spin and air conditions can alter where and when the ball reaches the goal. Using the comprehensive model's prediction for the given problem as a reference, we examine individual factors, alternative parameter combinations and simultaneous changes. Calculation details and parameter ranges are given in Appendix B.

5.1 Effects of Individual Factors

We first change one factor at a time and observe the resulting crossing position and arrival time. Table 5 reports the largest effects for the selected changes. Here D_g is the distance from the reference crossing, and Δt_g is the change in arrival time.

Table 5: One-parameter effects on goal-plane crossing and arrival time

Parameter	Perturbation	Maximum D_g (cm)	Maximum $\lvert\Delta t_g\rvert$ (ms)
Launch speed	$\pm 5\%$	8.356	20.659
Velocity azimuth	$\pm 1^\circ$	20.070	1.965
Velocity elevation	$\pm 1^\circ$	20.218	1.692
Spin magnitude	$\pm 5\%$	3.873	0.287
Spin-axis azimuth	$\pm 1^\circ$	1.002	0.071
Spin-axis elevation	$\pm 1^\circ$	0.852	0.084
Drag coefficient	$\pm 5\%$	0.620	2.274
Lift parameter	$\pm 5\%$	3.873	0.287
Air density	$\pm 5\%$	4.142	1.990

The initial direction has the greatest effect on the crossing: changing the horizontal or elevation angle by 1° shifts it by approximately 0.20 m. Speed has a pronounced effect on arrival time: a 5% change alters it by up to approximately 0.021 s. The direction determines where the ball travels, while speed affects how long gravity and aerodynamic forces act.

Greater spin generally produces more deflection. In this model, spin magnitude and the lift parameter jointly determine the rotational force; compensating changes in the two can produce the same trajectory. Air density affects both drag and Magnus force, thereby changing the ball's slowing and curvature. The effects on arrival time are shown in Figure 12.

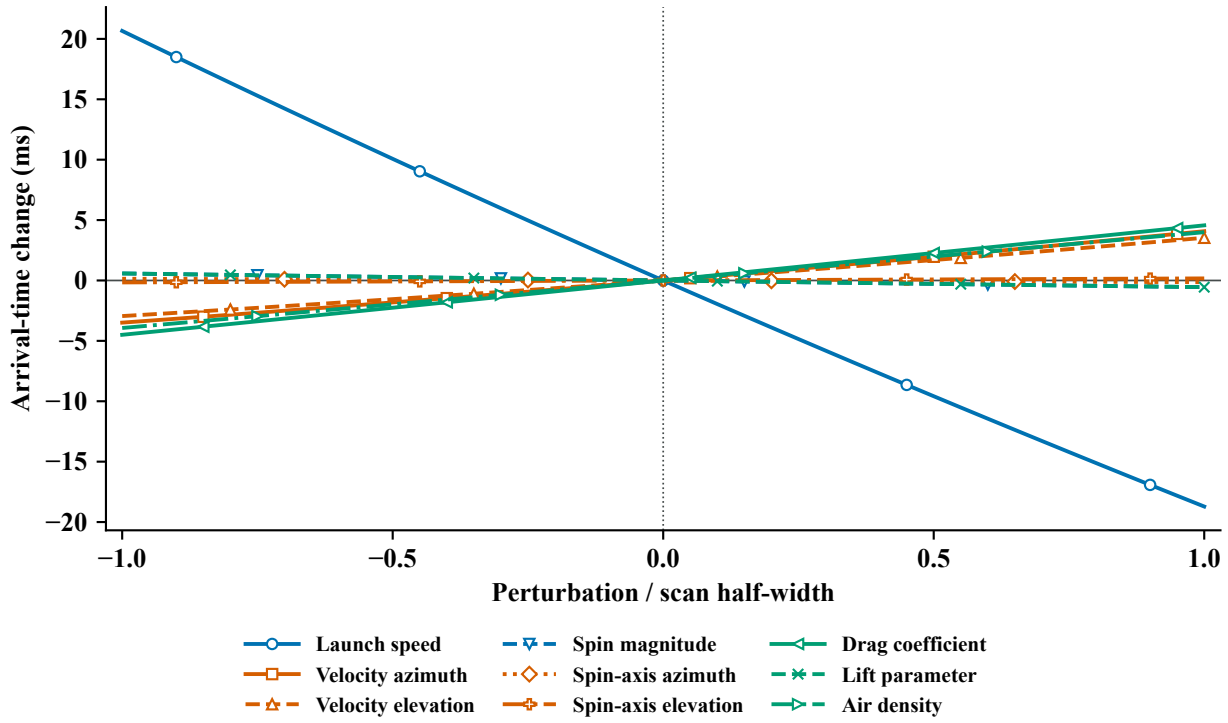


Figure 12: Arrival-time responses. Perturbation divided by scan half-width is plotted horizontally, so ± 1 denotes each scan endpoint. The vertical axis is Δt_g (ms); positive values mean later arrival.

5.2 Predictions with Similar Early Positions

Different combinations of initial velocity, spin and aerodynamic parameters can produce very similar early trajectories. How different might their goal-plane crossings be? We retain trajectories that closely match the given early positions and compare their crossings in Figure 13.

Within the specified parameter ranges, even when every early-position coordinate differs from its observation by no more than 2 mm, the largest crossing displacement found relative to the reference is approximately 0.107 m. This demonstrates uncertainty in predicting the goal-plane crossing from early positions alone.

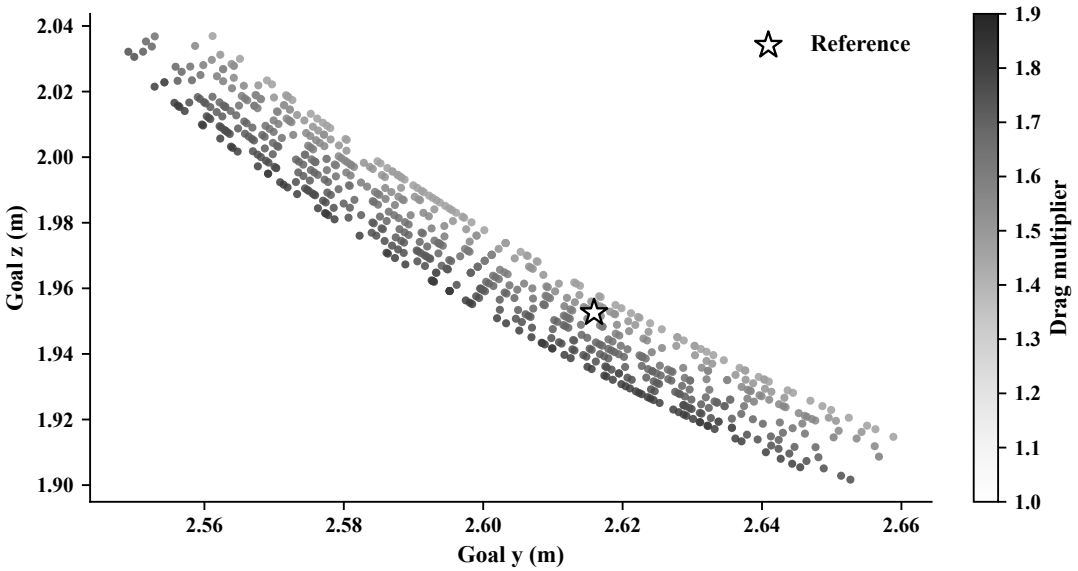


Figure 13: Goal-plane crossings satisfying the 2 mm early-position constraint. A sample of 700 points is shown; grey level denotes the drag multiplier and the star marks the reference crossing. The full search yielded 5049 crossings.

5.3 Combined Effects of Several Factors

In a real kick, speed, direction, spin and air conditions may change together. To examine their combined effect, we randomly generate 100000 sets of conditions within the specified ranges and calculate the position and time at the goal plane. The results appear in Table 6 and Figure 14.

Table 6: Crossings and arrival times for 100000 jointly perturbed trajectories

Output	Sample minimum–maximum	Central 95% interval, one coordinate
y_g (m)	1.9633–3.2143	2.1858–3.0505
z_g (m)	1.4381–2.4646	1.5534–2.3535
t_g (s)	0.3685–0.4346	0.3783–0.4218

The ellipse contains 95% of simulated crossings. Under the specified ranges and sampling procedure, 95.772% of trajectories pass completely through the goal, while the others cross above the crossbar. The distribution shows how a region of possible crossings can convey the variation in predictions alongside a single predicted position.

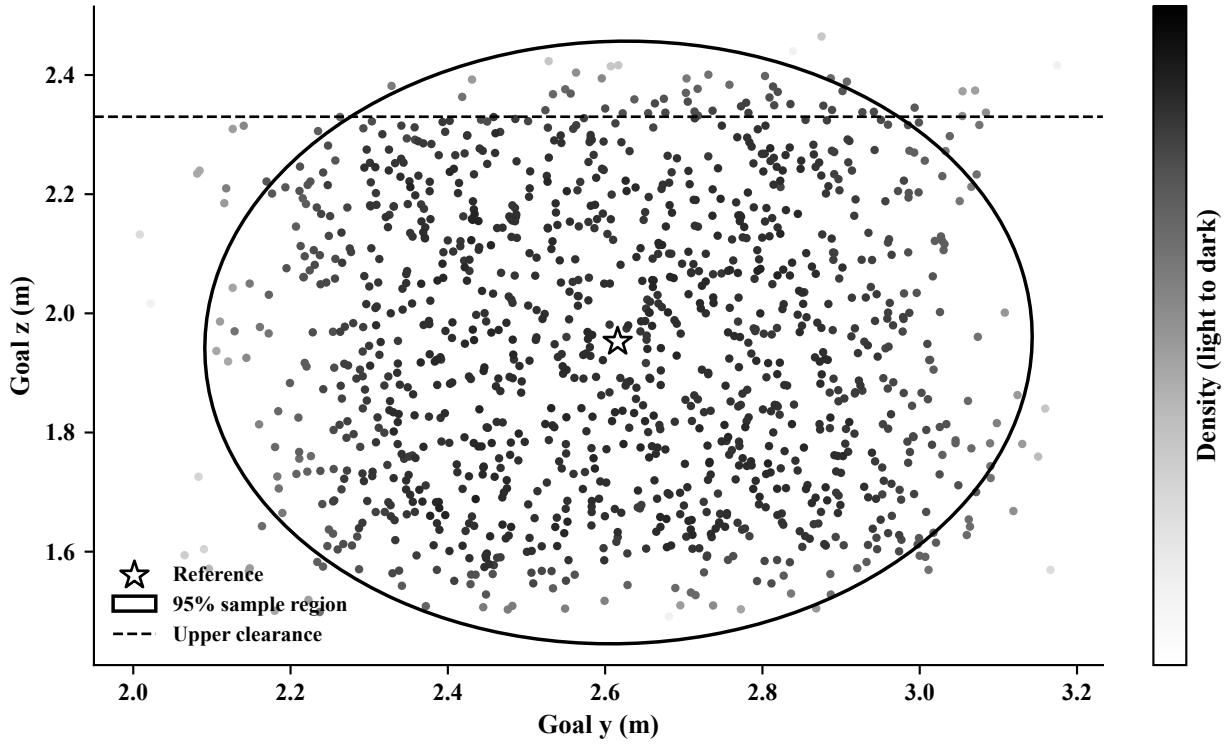


Figure 14: Goal-plane crossings under joint perturbations. A sample of 1200 points is shown, with darker shades indicating denser regions. The ellipse is determined from all 100000 simulations and contains 95% of their crossings; the dashed line marks the ball-centre height limit $z_g = 2.33$ m.

6 Discussion of Other Factors

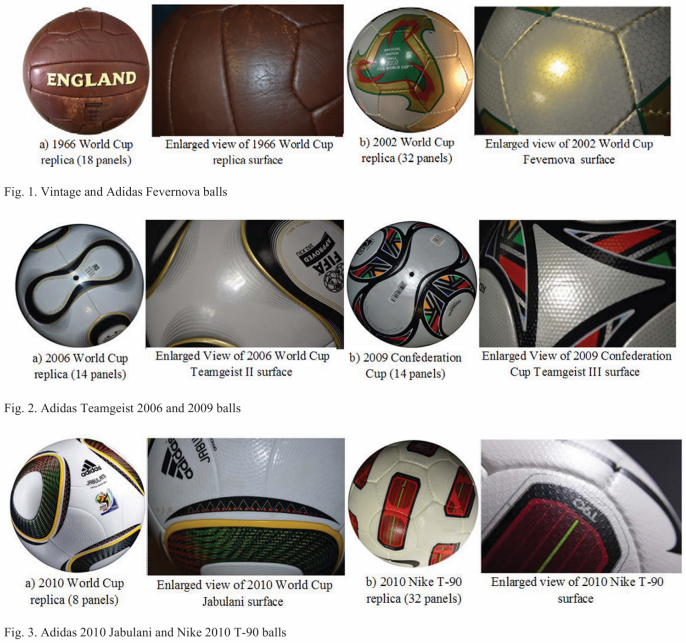
The main model isolates gravity, drag and Magnus lift. We now examine the physical effects omitted from its geometry, spin evolution and ambient airflow.

6.1 Asymmetric Surface Geometry

Full-flight simulations [4] distinguish realistic football geometry from a smooth sphere. As shown in Figure 15, the comparison includes both idealized geometry and actual panel textures; as shown in Figure 15a, we distinguish the seamed and smooth geometries. For launch speeds of 10, 15 and 20 m/s and spin rates of 31.5, 63 and 94.5 rad/s, their results show that seams and surface structure alter separation, wake dynamics and Magnus force. The smooth sphere produces greater lateral bending in their nine comparisons, and vertical flight-time differences become more visible at higher speed and rotation.



(a) Illustrative seamed-football and smooth-sphere comparison, following the comparison in [4].



(b) Six footballs and surface details, reproduced from Figures 1–3 of [10].

Figure 15: Football geometry, panels and surface textures.

Wind-tunnel measurements [10] reveal different drag coefficients for six footballs and a smooth sphere over a range of Reynolds numbers. As shown in Figure 15b, these footballs have different panels and textures; as shown in Figure 16, their drag coefficients also differ. The differences illustrate the influence of ball surface structure on aerodynamic drag.

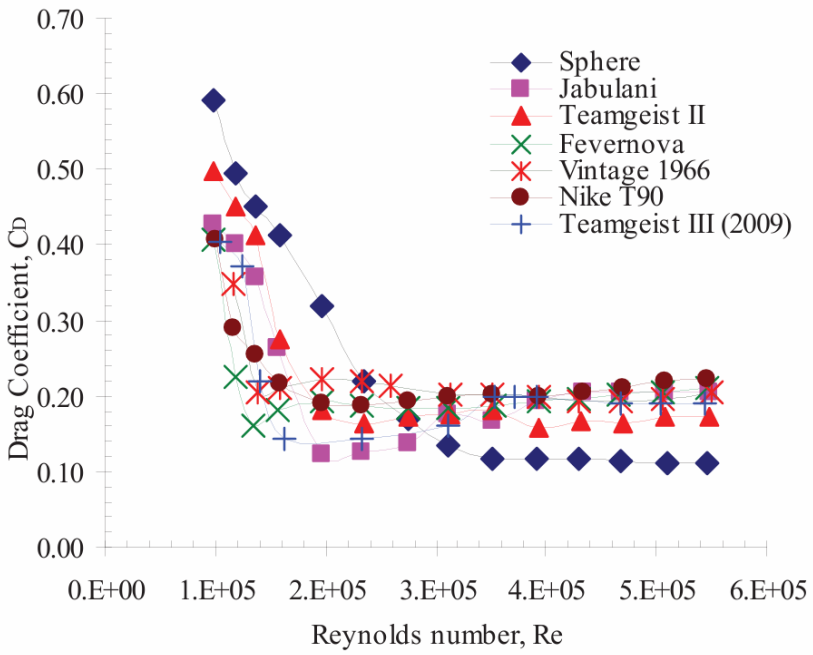


Figure 16: Drag coefficient against Reynolds number for different footballs and a smooth sphere, reproduced from Figure 5 of [10].

The spherical geometry in the present report describes the centre-of-mass motion and projected area. Its force coefficients come from football measurements, so the average surface influence enters through C_D and C_L. Explicit panel orientation and asymmetric wake fluctuations provide a further level of description when lateral-force details are important. These published results support retaining football-specific coefficients in a spherical trajectory model.

6.2 Rotation and Aerodynamic Damping

Aerodynamic torque opposes spin, gradually weakening rotation and changing the Magnus force. Following prior research and the MuJoCo model [7, 8], we use high-Reynolds-number rotational damping: stronger spin produces stronger resistance. The dimensionless angular-drag parameter $C_{\text{angular}} = 1.5$ is the software reference value. Following prior work [3], the inertia is approximated by that of a thin spherical shell. Viscous damping is much weaker than the inertial contribution in this example, so the comparison focuses on the latter.

The reference curved shot has initial velocity $(23.6354, 0, 4.1676)$ m/s, spin $(0, 0, 35)$ rad/s and unit force corrections. Holding this state fixed, we compare constant spin with rotational damping. As shown in Figure 17, decreasing spin slightly changes the flight curvature and goal-plane crossing.

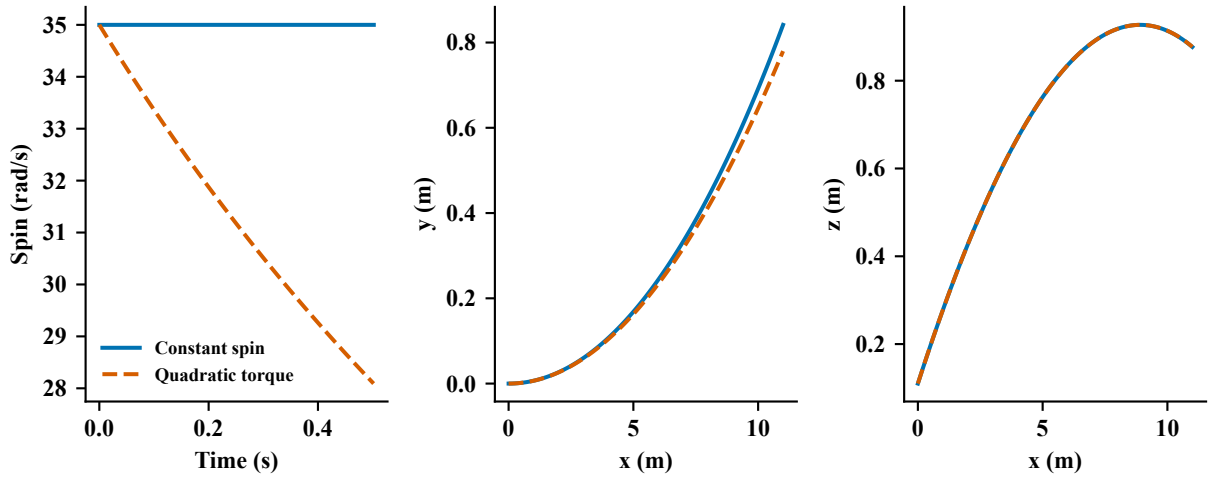


Figure 17: Constant spin and rotational damping under the same initial curved-shot conditions.

Spin falls by 19.740%, while the crossing moves 0.0616 m (0.560% of the 11 m distance) and arrival time changes by -0.066%. Rotational damping therefore has a small endpoint effect in this short reference flight.

6.3 Wind Speed and Force Estimates

Following an established relative-airflow treatment [3], we introduce air-relative velocity $\boldsymbol{u} = \boldsymbol{v} - \boldsymbol{v}_{\text{wind}}$ in this subsection, where $\boldsymbol{v}_{\text{wind}}$ is air velocity and $u = \|\boldsymbol{u}\|$ is relative speed. For wind, $\boldsymbol{u}$ and u replace $\boldsymbol{v}$ and v in the aerodynamic relations of Section 3. Let w be wind speed in m/s and v the ball speed. At fixed coefficients, denote the still-air dynamic pressure by q_0 and the headwind, tailwind and crosswind values by q_{head}, q_{tail} and q_{cross}. From the squared relative-speed dependence,

$$\frac{q_{\text{head}}}{q_0} = (1 + \frac{w}{v})^2, \qquad \frac{q_{\text{tail}}}{q_0} = (1 - \frac{w}{v})^2, \qquad \frac{q_{\text{cross}}}{q_0} = 1 + (\frac{w}{v})^2. \tag{11}$$

Crosswind rotates the airflow by $\arctan(\frac{w}{v})$ and changes the force direction. As shown in Table 7, illustrative winds of 1, 3 and 5 m/s alter dynamic pressure at the 24 m/s curved-shot speed. Let $F_{L,0}$, $F_{L,\text{head}}$ and $F_{L,\text{tail}}$ denote lift magnitudes in still air, headwind and tailwind. From Eq. (4), the M4 linear lift law depends on the first power of relative speed. With fixed spin and $0 \le w < v$,

$$\frac{F_{L,\text{head}}}{F_{L,0}} = 1 + \frac{w}{v}, \qquad \frac{F_{L,\text{tail}}}{F_{L,0}} = 1 - \frac{w}{v}. \tag{12}$$

Table 7: Illustrative wind at the 24 m/s curved-shot launch: dynamic-pressure changes at fixed coefficients.

w (m/s)	Headwind (%)	Tailwind (%)	Crosswind (%)	Angle (°)
1	+8.51	-8.16	+0.17	2.39
3	+26.56	-23.44	+1.56	7.13
5	+46.01	-37.33	+4.34	11.77

A 5% limit on the headwind dynamic-pressure change corresponds to $\frac{w}{v} \leq \sqrt{1.05} - 1$, or $w \leq 0.59$ m/s at this launch. The still-air approximation is most suitable for indoor or weak-wind flight; stronger wind should enter through the relative-velocity calculation.

7 Conclusions

We established and compared four motion models to predict a penalty-kick trajectory from the ball's initial position and early observations. The comprehensive model accounts for gravity, aerodynamic drag, and Magnus force, and predicts the subsequent trajectory after estimating the initial motion state and aerodynamic parameters. For the positions given in the problem, it predicts that the ball reaches the goal plane at approximately 0.399 s, with its center at $(11,\ 2.616,\ 1.953)$ m.

To evaluate predictions under different shooting conditions, we selected trajectories from a real penalty-kick dataset, using early positions as conditioning points and later positions as validation points. We examined accurate placement shots and powerful driven shots. Their initial speeds and spin states differ, as do the effects of drag and Magnus force on their trajectories. Comparison with measured trajectories shows that the comprehensive model captures the main features of the ball's flight.

Sensitivity analysis shows that the initial direction strongly affects the goal-plane crossing: a change of 1° shifts the crossing by approximately 0.20 m. Different initial motion states and aerodynamic parameters also produce a range of predicted crossing positions. In practice, assessing this range alongside a single predicted position can help a goalkeeper anticipate the direction of the incoming ball.

8 Model Evaluation

8.1 Model Strengths

Connecting forces and trajectory changes. The model incorporates gravitational fall, drag-induced deceleration and rotational deflection in a common equation of motion, relating trajectory curvature to changes in force magnitude and direction. The four-model comparison shows how different force descriptions and the constant-acceleration approximation affect predictions, providing a physical interpretation of the differences between low placement shots and higher power shots.

Connecting early inference and later validation. Three early positions are used to estimate initial motion and relevant parameters, after which the corresponding equations predict the subsequent trajectory. Later measured positions provide comparisons of the path and goal-plane crossing, connecting early fitting with subsequent predictive validation. Shared coordinates and definitions of ground contact and goal-plane crossing also make results comparable across models and shot cases.

Reference predictions and crossing ranges. Individual-parameter scans distinguish the effects on crossing position and arrival time. The position-constrained search reveals differences in predictions from parameter combinations matching the early observations, while joint perturbations give crossing distributions over specified ranges. Together, these analyses connect parameter changes to the final prediction, describing a reference crossing, response trends and distribution ranges.

8.2 Model Weaknesses

Coupling of initial motion and aerodynamic parameters. Over a short observation interval, initial velocity, spin and aerodynamic parameters can compensate for one another, allowing initially similar trajectories to separate later in flight. In the linear lift relation, spin magnitude and the lift parameter enter the force through their product, so different parameter values can also generate the same trajectory. Additional early tracking positions and direct spin measurements would further constrain initial motion and aerodynamic parameters.

Simplified aerodynamics and environmental conditions. The main model treats the drag coefficient and linear lift parameter as constants during each flight and assumes still air and constant spin. Flow transitions, ball orientation and wake fluctuations alter aerodynamic forces, while wind and rotational damping affect relative airflow and spin evolution. More detailed descriptions require aerodynamic data for each ball type, together with wind and spin measurements.

Limited coverage of measured conditions. The measured comparison covers two placement shots and one power shot. Tracking records spanning additional ball types, shot techniques and wind conditions would allow assessment across more operating conditions. Joint perturbations use independent uniform sampling of nine parameters; estimating their distributions and correlations from measurements would bring the simulated crossing distribution closer to actual shooting conditions.

References

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Appendix A: Data Sources, Frame Selection and Parameter Bounds

Following the dataset description by Bassek et al. [1] and the DFL release [2], we retain the source event and frame identifiers. The three selected penalties contain 39 records, with 15, 12 and 12 for the respective events. As shown in Table 8, each case is associated with its player and selected samples.

Table 8: Correspondence between case labels and source records

Case	Player	Selected samples
Case 1	Dejan Ljubicic	3, 4, 5
Case 2	Philipp Klement	3, 4, 5
Case 3	Dawid Igor Kownacki	2, 3, 4

The first selected position is the time origin and the fitting times are 0, 0.04 and 0.08 s. Their source-time origins are 0.08, 0.08 and 0.04 s in the table's order. Repeated and post-goal records are retained; later positions are used only for validation. The observed goal crossing uses a source frame or interpolation. Let $(X, Y, Z_{\text{original}})$ denote the published pitch-centred coordinates. The height is interpreted in metres as ball-centre height, and the local coordinates follow from $x = X - 41.5$, $y = Y$ and $z = Z_{\text{original}}$.

As shown in Table 2, the retained inputs and observed crossings share one local coordinate system. As shown in Table 9, the inverse variables have specified search ranges.

Table 9: Component bounds and force-correction settings for inversion.

Quantity	Search interval
v_{x0} (M1, M2, M4)	$[0, 35]$ m/s
v_{y0}, v_{z0} (M1, M2, M4)	$[-35, 35]$ m/s
$\|\boldsymbol{\omega}_0\|$ (M4)	30 rad/s
Azimuth, elevation (M4)	$[-\pi, \pi], [-\frac{\pi}{2}, \frac{\pi}{2}]$ rad
$\frac{C_D}{C_D^*}, \frac{\alpha}{\alpha^*}$	$[0.5, 1.5]$

As shown in Table 10, the M4 states differ in speed, coefficients and unit spin axis $\boldsymbol{n} = \dfrac{\boldsymbol{\omega}}{\|\boldsymbol{\omega}\|}$.

Table 10: Fitted M4 states at the selected starting frames.

Event	v_0 (m/s)	C_D	α	n_x	n_y	n_z
Case 1	29.20	0.3623	2.2877	0.2280	-0.9728	0.0413
Case 2	28.34	0.1208	2.2877	0.1439	-0.2711	-0.9517
Case 3	32.05	0.3238	2.2877	-0.2276	0.3170	0.9207

Reproduction Code

The following excerpts evaluate the force components and state derivatives used in Eq. (7). Initial-state inference and subsequent trajectory prediction use these same physical relations and coefficient settings.

```
def state_derivative(t, state, config):
    """The single nine-state RHS used for prediction and inverse fitting."""
    f = forces(state[3:6], state[6:9], config)
    return np.r_[state[3:6], sum(f.values())/config.mass,
                 rotational_torque(state[6:9], config)/config.moment_of_inertia]
```

```
def forces(velocity, spin, config):
    """Return force components on ball, in newtons."""
    v = np.asarray(velocity, dtype=float) - np.asarray(config.wind, dtype=float)
    omega = np.asarray(spin, dtype=float)
    speed = np.linalg.norm(v)
    weight = (config.mass * np.asarray(config.constant_acceleration, dtype=float)
              if config.constant_acceleration is not None else
              np.array([0., 0., -config.mass * config.gravity if config.use_gravity else
                  0.]))
    drag = np.zeros(3)
    magnus = np.zeros(3)
    if speed > 1e-12:
        cd, cl, _, _, _ = coefficients(velocity, spin, config)
        q_area = 0.5 * config.air_density * np.pi * config.radius**2 * speed**2
        if config.use_drag:
            drag = -q_area * config.drag_scale * cd * v / speed
        cross = np.cross(omega, v)
        norm = np.linalg.norm(cross)
        if config.use_magnus and norm > 1e-12:
            # The literature fit is for transverse spin. Project its effect
            # onto the plane perpendicular to airflow for an oblique spin axis.
            sin_angle = norm / (np.linalg.norm(omega) * speed)
            magnus = q_area * config.lift_scale * cl * sin_angle * cross / norm
    return {"gravity": weight, "drag": drag, "magnus": magnus}
```

Appendix B: Sensitivity Analysis Details

This chapter uses the M4 state inferred in Section 4.3 to examine how initial motion, aerodynamic parameters and air density affect the goal-plane crossing and arrival time. The model includes gravity, drag and Magnus force, with still air, constant spin, a fixed mass of 0.43 kg and a fixed radius of 0.11 m. The early positions in Table 4 also constrain the alternative trajectories considered below.

The reference crossing is $(y_g^*, z_g^*) = (2.6159, 1.9526)$ m, with arrival time $t_g^* = 0.398967$ s. The corresponding coefficients are $C_D = 0.3623$ and $\alpha = 2.2877$, and the maximum early-coordinate residual is 0.784 mm.

We first vary one parameter at a time to identify its effect on the crossing and arrival time, then retain the early-position constraints to compare predictions from different parameter combinations. Finally, joint perturbations generate the crossing distribution and its statistical ranges.

Effects of Individual Parameters on Crossing Position and Arrival Time

Each parameter is varied separately while the others remain at their reference values. Table 11 lists the nine perturbation ranges: percentages denote changes relative to the reference value, and direction angles vary about their reference orientation. Each parameter takes 41 equally spaced values, giving 369 trajectories.

Table 11: Perturbation ranges for the nine parameters

Parameter	Perturbation range
Launch speed	$\pm 5\%$
Velocity azimuth	$\pm 2^\circ$
Velocity elevation	$\pm 2^\circ$
Spin magnitude	$\pm 10\%$
Spin-axis azimuth	$\pm 2^\circ$
Spin-axis elevation	$\pm 2^\circ$
Drag coefficient	$\pm 10\%$
Lift parameter	$\pm 10\%$
Air density	$\pm 10\%$

The changes in crossing coordinates and arrival time relative to the reference trajectory are defined as

$$\begin{aligned} \Delta y_g &= y_g - y_g^*, \qquad \Delta z_g = z_g - z_g^*, \qquad \Delta t_g = t_g - t_g^*, \\ D_g &= \sqrt{(\Delta y_g)^2 + (\Delta z_g)^2}. \end{aligned} \tag{13}$$

Figure 12 shows the arrival-time responses. Table 5 takes direction perturbations of $\pm 1^\circ$ and other perturbations of $\pm 5\%$ from these scans, reporting the larger displacement and absolute time change from each positive–negative pair.

For the perturbation sizes in Table 5, initial direction primarily affects crossing position: a 1° change in azimuth or elevation moves the crossing by about 0.20 m. Speed strongly affects arrival time; a $\pm 5\%$ change produces a maximum time difference of 20.659 ms and a maximum crossing displacement of 8.356 cm. Direction directly determines lateral and vertical motion, while speed changes the time to the goal and therefore the accumulated effects of gravity and aerodynamic forces.

The responses to spin magnitude and the lift parameter coincide because the linear Magnus force is proportional to the product $\alpha\boldsymbol{\omega}$. The same relative change in either spin magnitude or α produces the same force and trajectory response. Scaling them inversely gives

$$\boldsymbol{\omega}_0' = k\boldsymbol{\omega}_0, \qquad \alpha' = \alpha/k \quad \Longrightarrow \quad \boldsymbol{F}_L' = \boldsymbol{F}_L, \qquad k > 0. \tag{14}$$

Different spin magnitudes and lift parameters can therefore produce the same trajectory. Spin-axis orientation affects the magnitude and direction of deflection through $\boldsymbol{\omega} \times \boldsymbol{v}$. Air density scales both drag and Magnus force, changing deceleration and curvature together.

Prediction Differences under Early-Position Constraints

The preceding analysis varies one parameter while holding the others fixed. We now examine how much the goal-plane crossing can change when different parameter combinations reproduce the same early positions approximately.

Using the aerodynamic reference values C_D^* and α^* from Section 4, define

$$\lambda_D = C_D/C_D^*, \qquad \lambda_\alpha = \alpha/\alpha^*. \tag{15}$$

Each coefficient multiplier takes 17 equally spaced values in $[1.2, 1.8]$, forming 17×17 combinations. For each combination, the initial spin component ω_{x0} takes 33 equally spaced values in $[-60, 60]$ rad/s. The three initial velocity components and the other two spin components are then fitted to the early positions in Table 4. The constraints are a total speed not exceeding 35 m/s, an absolute value not exceeding 60 rad/s for each spin component, and a maximum early-coordinate residual not exceeding 2 mm.

The supplementary search yields 5049 compatible trajectories across 153 aerodynamic-parameter combinations. Restricting both multipliers to $[1.35, 1.65]$, corresponding to $\pm 10\%$ changes in each aerodynamic parameter about its reference-trajectory value, gives 2079 trajectories across 63 combinations. Figure 13 shows their crossing positions.

Within the wider parameter range, the largest crossing displacement among the searched trajectories is 0.1074 m relative to the reference. The minimum-residual solutions for each aerodynamic-parameter combination lie at the spin-component bounds; plotting these solutions alone produces two clusters. Intermediate spin states satisfying the same 2 mm position constraint form the band joining the clusters in Figure 13. Thus, initial velocity and spin can compensate for aerodynamic changes over the early observation interval, while different force histories accumulate into distinct subsequent crossings.

Crossing Distribution under Joint Perturbations

To examine simultaneous changes in several conditions, the nine parameters are sampled independently and uniformly over the ranges in Section 5.1, producing 100000 inputs. Mass, radius and initial position remain fixed. Integrating the equations for each input gives the crossing distribution under jointly varying launch conditions and aerodynamic parameters.

The central 95% intervals in Table 6 use the 2.5th and 97.5th percentiles of each output. The mean crossing is $(2.6170, 1.9511)$ m, with lateral and vertical standard deviations of 0.2455 m and 0.2360 m. The largest displacement from the reference crossing is 0.801 m.

To describe lateral and vertical variation jointly, write $\boldsymbol{q}_i = (y_{g,i}, z_{g,i})^{\mathsf{T}}$ and use the sample mean $\bar{\boldsymbol{q}}$ and covariance matrix $\widehat{\Sigma}$ to define

$$d_i^2 = (\boldsymbol{q}_i - \bar{\boldsymbol{q}})^{\mathsf{T}} \widehat{\Sigma}^{-1} (\boldsymbol{q}_i - \bar{\boldsymbol{q}}). \tag{16}$$

Taking the empirical 95th percentile $c_{0.95} = 4.5916$ of d_i^2 gives the joint region

$$\mathcal{E}_{0.95} = \left\{ \boldsymbol{q} : (\boldsymbol{q} - \bar{\boldsymbol{q}})^{\mathsf{T}} \widehat{\Sigma}^{-1} (\boldsymbol{q} - \bar{\boldsymbol{q}}) \le c_{0.95} \right\}. \tag{17}$$

The set $\mathcal{E}_{0.95}$ is the ellipse boundary and its interior in Figure 14. The ellipse line marks the boundary, and the region contains 95% of the simulated crossings. The goal is 2.44 m high; accounting for the 0.11 m ball radius requires a centre height between 0.11 and 2.33 m and a lateral position satisfying $|y_g| \le 3.55$ m for the whole ball to pass through. By these criteria, 95772 trajectories clear the opening, or 95.772%; the other 4228 trajectories, or 4.228%, exceed the 2.33 m centre-height limit.

Figure 18 displays 400 randomly selected trajectories, linking the nine inputs to crossing coordinates and arrival time. Inputs are scaled to 0–1 using their perturbation bounds; outputs use the extrema of the full 100000-trajectory sample. The subset contains 385 blue lines for trajectories clearing the goal and 15 orange lines for trajectories crossing too high. Each line connects one set of launch conditions and aerodynamic parameters to its predicted outcome.

Figure 19 checks whether the crossing statistics stabilize as the number of simulations increases, informing the choice of 100000 samples for estimating the central 95% intervals. The blue sample extrema expand as additional extreme trajectories appear, while the orange percentile intervals gradually stabilize. Increasing the sample from 50000 to 100000 changes the lateral and vertical interval endpoints by at most 0.264 mm and 0.506 mm, and the arrival-time endpoints by at most 0.138 ms. Two additional runs of 10000 trajectories with independent random seeds differ from the main sample by at most 5.425 mm laterally, 4.320 mm vertically and 0.328 ms in time. These comparisons show that further sampling has a small effect on the central-interval estimates.

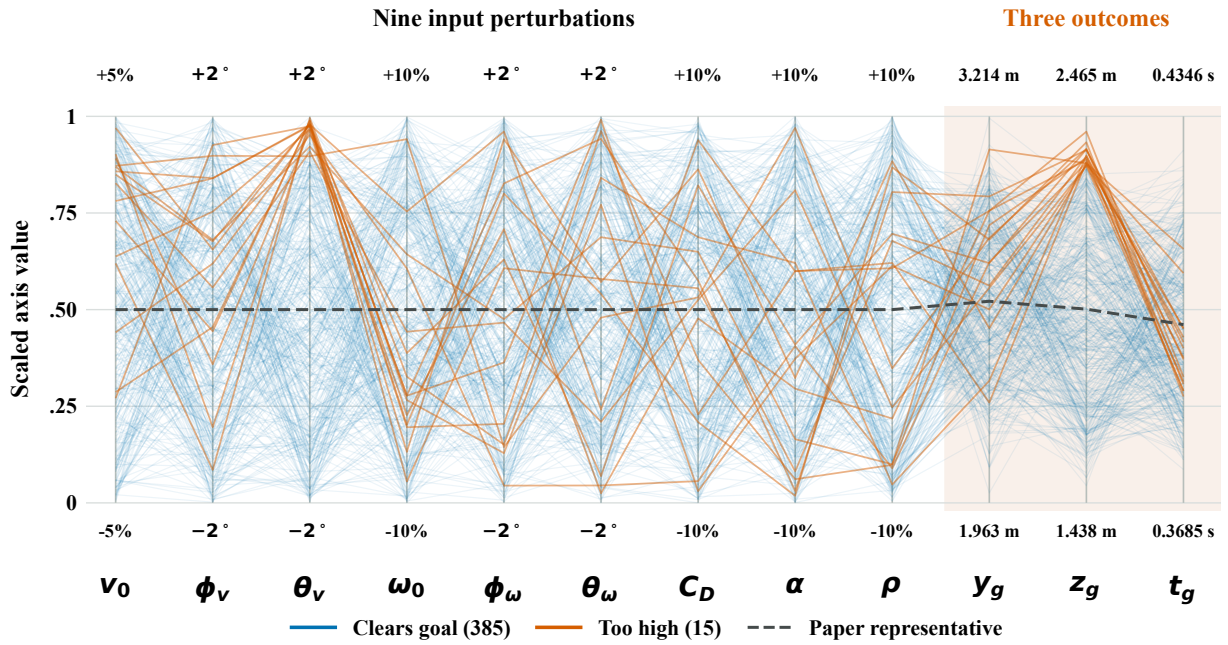


Figure 18: Parallel coordinates linking perturbed inputs, crossing coordinates and arrival time. The first nine axes are inputs and the last three are outputs. The displayed subset contains 385 trajectories clearing the opening and 15 crossing too high.

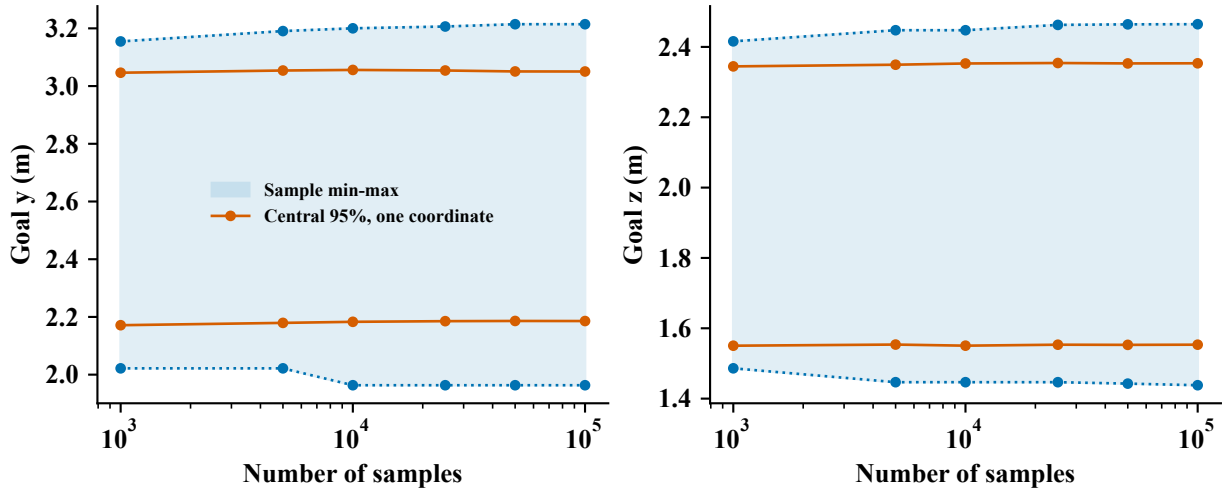


Figure 19: Convergence of crossing statistics with simulation sample size. Blue shading and dotted curves show sample extrema; orange curves show each coordinate's central 95% interval, illustrating the stability of interval estimates as sample size increases.